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LETTER

A New BBR with Moderate Delivery Rate

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Abstract BBR has been proposed as congestion-based congestion control by Google. It was originally designed to operate around optimal operating point where maximum throughput is obtained without incurring increase of roundtrip delay. In this paper, with simple analysis and detailed simulation results, we show that BBR increases its delivery rate even though BBR flows sharing the same bottleneck link fully utilize available bandwidth. From simple analysis results, we propose a new BBR delivery rate calculation algorithm which regulates delivery rate moderately, BMR. Our simulation results show that our proposed BMR operates around the optimal operating point.

Keywords: Congestion Control, BBR
Classification: Internet

1. Introduction

As available network bandwidth in the current Internet has increased, routers are increasingly designed with large buffer sizes. With loss-based TCP, such as CUBIC [1], router buffer is fully occupied at the timing of congestion detection at the end host. Large queue size at the intermediate router on the end-to-end path causes large end-to-end delay, which is a well-known technical problem of "bufferbloat" [2].

Google proposed a new type of congestion control, congestion-based congestion control, BBR(Bottleneck Bandwidth and Round-trip-time) in 2016 [3], which is expected to resolve bufferbloat. BBR aims to operate at the optimal operating point proposed by Kleinrock [4](for details, see Sec.2). BBR is reported to be second place in deployment of TCP variants and accounts for more than 40% volume of the Internet traffic [5]. Even though BBR is originally designed to operate at the optimal operating point, M.Hock et al. [6] claim that BBR continues to increase its delivery rate even after total rate excesses bottleneck bandwidth. They also analyze delivery rate of BBR and reveal that when gain cycles of BBR flows are asynchronous, BBR inflight increases until it reaches inflight cap defined as twice of BDP (Bandwidth Delay Product). This means BBR which is originally designed to resolve bufferbloat, surprisingly might cause bufferbloat.

In this paper, we simply analyze BBR delivery rate just after seek phase (gain = 1.25) [7]. In this analysis, we first assume that two flows share the bottleneck link fairly. Then, we analyze estimated available bandwidth for this flow just after the seek phase and reveal that it increases 10/9th higher. These analysis results also show that BBR flows increases their delivery rates even though their total rate reaches bottleneck bandwidth. Injected packets over bottleneck bandwidth are stored at the bottleneck router, which finally cases bufferbloat. Based on our analysis results, in this paper, we propose a new BBR algorithm which regulates its delivery rate moderately, BBR with Moderate delivery Rate (BMR).

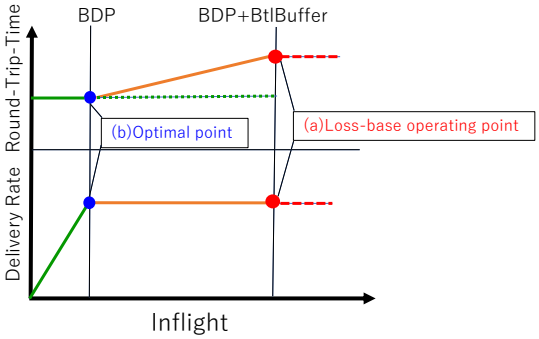


Fig. 1: Operating point of congestion control

Our contributions in this paper are, (1) with simple analysis [7], we show that delivery rate of BBR increases ten ninths(10/9) higher just after seek phase even though bottleneck bandwidth is fully utilized, (2) we evaluate detailed performance of BBR's delivery rate which increases just after seek phase by computer simulation(M.Hock et al. [6] just show experimental evaluation results without detailed ones), (3) based on these analysis results, we propose a new BBR algorithm, BMR, which resolves bufferbloat in BBR.

2. BBR

BBR is designed to operate around the optimal operating point proposed by Kleinrock [4] which is depicted as blue points in Fig.1. With increase of inflight packets, until it reaches BDP, delivery rate increases linearly. RTT has fixed value of propagation delay because no packets are stored in the bottleneck router. After it reaches BDP, delivery rate does not increase and inflight packets over BDP is stored at the bottleneck router. So, RTT increases linearly with increase of queueing delay. At the optimal operating point, the maximum delivery rate can be obtained with minimum RTT.

BBR is congestion-based congestion control because it is designed to operate at the optimal operating point where maximum throughput can be obtained without congestion.

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Because available network bandwidth dynamically changes, BBR needs to search the "current" optimal operating point. For this purpose, BBR measures two parameters $RTprop$, minimum filtered RTT for 10seconds, and $BtlBw$, maximum filtered delivery rate for 10RTprop. At every ACK reception, BBR measures the amount of acknowledged data volume for 1RTT. And delivery rate is calculated as average acknowledged data volume in 1RTT. $BtlBw$ is the maximum delivery rate over 10RTprop.

The sending rate of BBR is calculated by $BtlBw \times gain$. By increasing or decreasing gain, BBR can search maximum filtered delivery rate without excess queueing. The gain is decided through the following BBR four states.

- Startup: Startup is initial state quickly adjusting sending rate to available bandwidth. In this state, gain is set to 2/ln2 and rate increases exponentially like slow-start phase in TCP. Without increase of $BtlBw$ in 3RTTs, upper-bound bandwidth might be obtained. So, BBR exits this Startup phase and enters the next phase, "Drain".
- Drain: This state is for eliminating packets in the bottleneck queue built at the Startup phase, and pacing gain is the inverse of the value used at Start up(=ln2/2).
- probeBW: probeBW is a steady state of BBR. The aim of this phase is a search for further available bandwidth. The gain is changed in a cyclic fashion and has 8 cycles of gaincycle=$\{1.25,0.75,1,1,1,1,1,1\}$. Each gain-term basically continues for RTprop. The seek cycle of gain=1.25 is for searching more available bandwidth, and the drain cycle of gain=0.75 is for absorbing packets accumulated in the queue in the seek cycle. In a cycle with gain=1, delivery rate is just maintained.
- probeRTT: The probeRTT phase is also one of two BBR steady states. When RTprop has not been updated for 10 seconds, BBR enters this phase. cwnd(congestion window size) is reduced to 4 and held for 200 ms to completely absorb the queue at the bottleneck router. By removing packets in the router buffer, the minimum RTT, RTprop, can be correctly re-measured. As described in the original paper of BBR [3], when multiple BBR flows share the same bottleneck link, the probeRTT phase is synchronized for these flows. After 200ms have passed in the probeRTT phase, BBR returns to the probeBW phase.

BBR is basically rate-based, but to avoid injecting the excessive data in the network, the upper limit of inflight, called inflight cap, is set to 2BDP.

3. Simple analysis and Preliminary evaluation of BBR

3.1 Simple analysis

In this paper, we first simply analyze $BtlBw$ behavior in the probeBW phase. The seek cycle (gain = 1.25) plays an important role for increase of $BtlBw$. So, we focus on the last packet sent in the seek cycle and induce mathematical equations in terms of the amount of data acknowledged during the round-trip delay of this focused packet.

We first assume that two flows A and B with equal RTT (=$RTprop$) share the same bottleneck link. We

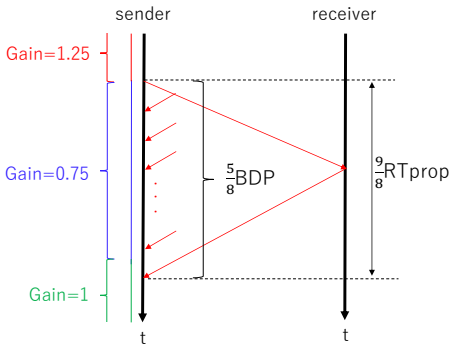


Fig. 2: Time sequence of gain cycles

calculate the transition from a situation where both flows have a fair allocation of bottleneck link bandwidth ($BtlBw_A$=$BtlBw_B$=$\frac{Bw}{2}$) and there is no stored packets in the bottleneck router. In BBR, the initial gain cycle is randomly selected from other than the drain cyble(gain = 0.75). In other words, since the gain transition is asynchronous for each flow with high probability, in this analysis, we calculate the transition from the state where the gain cycles of flows A and B are 1.25 and 1, respectively.

The interval when the sending rate of flow A is $1.25 \times BtlBw_A$ lasts for the RTprop. That is, the amount of data sent by Flow A during this period is obtained as follows.

$$\frac{5}{4} BtlBw_A \times RTprop = \frac{5}{8} BDP, \tag{1}$$

where BDP=$Bw \times RTprop$. The amount of data sent in this interval by flow B is $\frac{BDP}{2}$, and $\frac{9}{8}BDP$ is totally sent from both flows within this interval.

Now, we would like to induce the amount of data acknowledged by ACKs within the round-trip delay(amount of data acknowledged by the red arrow ACKs in Fig.2) of our focusing last packet of the seek phase. The round-trip delay is equal to the delay in which the above total data volume is transferred by the bottleneck bandwidth, Bw,

$$\frac{\frac{9}{8}BDP}{Bw} = \frac{9}{8} RTprop. \tag{2}$$

By equation (1) and (2), new $BtlBw_A$ which is max-filtered with newly obtained available bandwidth estimation, is slightly increased as follows.

$$\frac{\frac{5}{8}BDP}{\frac{9}{8}RTprop} = \frac{5}{9} Bw > BtlBw_A = \frac{Bw}{2}. \tag{3}$$

Before entering the seek cycle, available delivery rate of flow A, $BtlBw_A$, is $Bw/2$. After entering the seek cycle, it increases continuously and finally just after reception of the ACK of the last packet in the seek cycle, it is $5Bw/9$(as shown in equation 3). This means for flow A, its available delivery rate increases ten ninths(10/9) higher, as follows.

$$\frac{\frac{5}{9}Bw}{\frac{Bw}{2}} = \frac{10}{9}. \tag{4}$$

Thus, $BtlBw_A$ estimated in the seek cycle(gain=1.25) is slightly increased even though the seek cycle starts in the

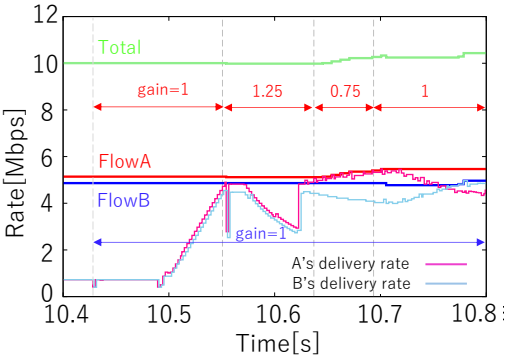


Fig. 3: $BtlBw$ and delivery rate of two BBR flows (short-term range)

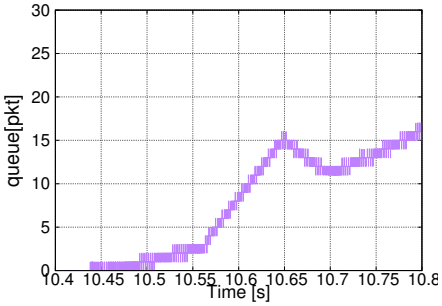


Fig. 4: Bottleneck queue length (short-term range)

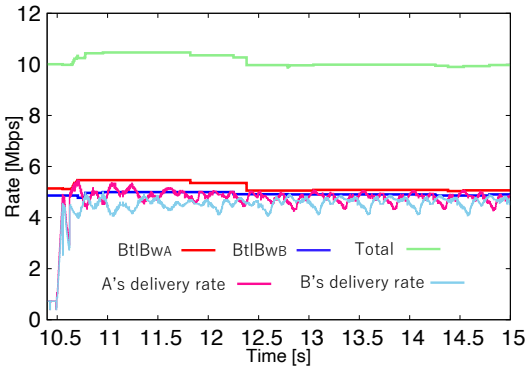


Fig. 5: $BtlBw$ and delivery rate of two BBR flows (long-term range)

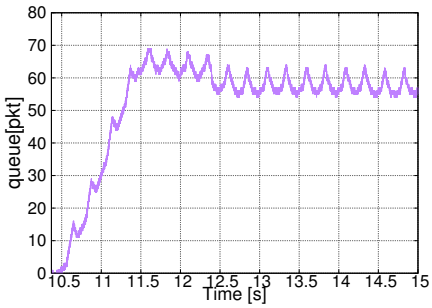


Fig. 6: Bottleneck queue length (long-term range)

situation that two flows just fully utilize the bottleneck link, i.e. $BtlBw_A + BtlBw_B = Bw$. This leads to the increase in the amount of inflight. Finally, inflight might reach the inflight cap, and after that BBR operates at inflight cap.

The reason why the measured delivery rate increases through the seek cycle is that total volume of data packets increases with the seek gain(1.25) and on the other hand, round-trip delay is not so much increased(for example, in our simple analysis case, nine eighths higher). The reason of smaller increase of round-trip delay is the other flow does not increase its delivery rate(due to gain=1) and traffic load to the bottleneck link does not increase as high as the focused flow(traffic volume of the focused flow increases one and quarter times). This means asynchronous gain cycle is the root cause of measured delivery rate increase.

3.2 Preliminary evaluation

We evaluate detailed BBR behavior around the seek phase (gain=1.25) by computer simulation using ns3. In our simulation model, RTT is 60[ms], Bw is 10[Mbps], and packet size is 1500[bytes]. BDP=$\frac{10\times10^6\times0.06}{8\times1500}$ =50[pkt].

Figure 3 shows measured data delivery rates (light red and blue lines) and max-filtered $BtlBw$(red and blue lines) of the two flows. In this simulation experiment, probeRTT phase ends at 10.43s. And after eliminating the buffer in the probeRTT phase, the probeBW phase starts and both flows share bandwidth fairly with gain=1(10.43-10.55s), then at 10.55s flow A starts the seek cycle(gain=1.25). After entering the drain cycle (gain=0.75), $BtlBw_A$ is updated due to the increase in measured data delivery rate($BtlBw_A$ is max-filtered), indicating that the sum of $BtlBw$(light green line) exceeds Bw (10Mbps).

Figures 4 and 6 show queue length at the bottleneck router. The former figure shows short-term characteristics(10.4-10.8s) and the latter shows long-term one(10.4-15s). As shown in Fig. 4, queue length grows during the seek cycle(until 10.65) and slightly decreases during the drain cycle. And it grows constantly after the drain cycle. This is because total delivery rate of the two flows exceeds the bottleneck bandwidth. As shown in Fig. 6 queue length continuously increases until BBR reaches the inflight cap.

The cause of this increase in the bottleneck queue length is slight increase of measured data delivery rate, and this tentative increase of the measured data delivery rate(as shown in Fig. 3, measured data delivery rate of flow A decreases just after it has tentative maximum value) is maintained for 10RTTs by the max-filter. Figure 5 shows long-term measured data delivery rates and max filtered $BtlBw$ of the two flows. After decrease of measured data delivery rate of flow A(just after 10.7s as shown in Fig. 3), measured data delivery rate of flow A tentatively increases several times with the seek cycles. As shown in Fig. 5, for example, at 10.9s, 11.1s, 11.3s and so on, measured data delivery rate has tentatively maximum values. These slight increases of measured data delivery rate cause higher value of max-filtered $BtlBw_A$ beeing kept for longer periods (than 10RTTs). This mechanism prevents BBR from operating at the optimal operating point. From these analysis and simulation results, we came up with the idea that the delivery rate of BBR is to be set

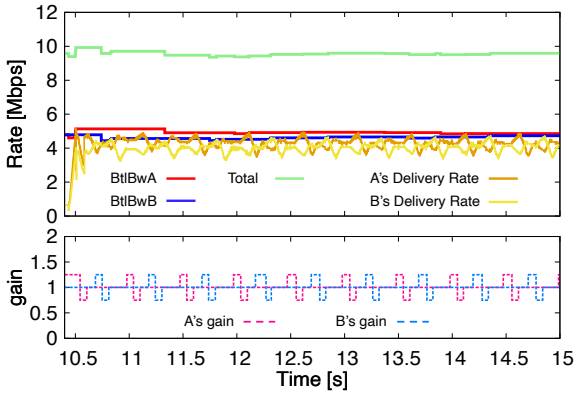


Fig. 7: $BtlBw$ and delivery rate of BMR(α=0.9)

slightly lower than max-filtered $BtlBw$.

4. BMR:BBR with Moderate delivery Rate

As described above, we would like to propose a new delivery rate calculation algorithm for BBR, BMR(BBR with Moderate delivery Rate). As shown in our simple analysis results, tentative delivery rate just after the seek phase increases $\frac{1}{9}$ of the former value when the flow exits its seek cycle. We would like to prevent the transmission rate of BBR from excessing the bottleneck bandwidth. So, we propose a simple method of setting the transmission rate of BBR to $\alpha \times BtlBw$, where α should be less than or equal to $\frac{9}{10}$.

We evaluate our proposed BMR with the same simulation model used in chapter 3. Figure 7 shows $BtlBw$, measured data delivery rate and gain cycles of BMR for the case of α = 0.9. Light green line shows sum of $BtlBw_A$ and $BtlBw_B$. And Fig. 8 shows queue length of BMR for α = 0.9. As shown in Fig. 7, gain cycles of two flows are not synchronized. As we discussed in section 3.1, asynchrony of gain cycle causes increase of measured delivery rate of BBR. Measured delivery rates of flow A and B increase just after their seek cycle. And $BtlBw_A$ and $BtlBw_B$ are max-filtered with these delivery rates. In our proposed BMR, transmission rate of each flow is moderately regulated to 0.9-times $BtlBw_A$ and $BtlBw_B$ for flow A and B, recpectively.[1] With this moderate regulation to transmission rate, max-filtered measured delivery rate, $BtlBw_A$ and $BtlBw_B$ is stable without any increase. And total transmission rate is 0.9-times ($BtlBw_A$ + $BtlBw_B$), i.e., it is slightly lower than 10Mbps(0.9-times higher value of the light green line is slightly below 10Mbps). Because of slight lower transmission rate than 10Mbps bandwidth, queue length is also stable without any increase(of course, there are some spike due to the seek cycle).

As total delivery rate is close to Bw(although slightly smaller than Bw due to weight of α) and queue length is not reduced to 0, bottleneck link is ideally utilized without bufferbloat. This means our proposed BMR can operate around the optimal operating point which is the original goal of BBR. From queueing theory, full utilization is quite risky for stable system behavior and this is also the case for BBR congestion control. We believe that our proposed BMR which moderately regulates transmission rate just below the full utilization, is quite a reasonable approach.

[1] Exactly, for flow A, transmission rate is $BtlBw_A \times gain \times \alpha$. In our discussion here, "gain" is omitted for simple and easy explanations.

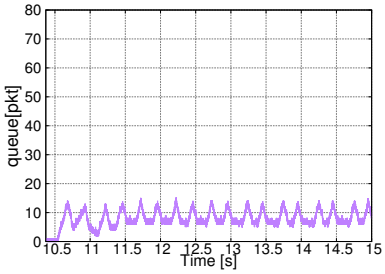


Fig. 8: Bottleneck queue length of BMR(α=0.9)

5. Conclusion

In this paper, we focus on intra-protocol behaviors of BBR congestion control. Our simple analysis and preliminary evaluation results show that although BBR is designed to operate at the optimal operating point, BBR is operating at inflight cap, not at the optimal operating point. And our simple analysis reveals that delivery rate of a flow increases $\frac{10}{9}$ higher just after the seek cycle in the ProbeBw phase of BBR. Based on these results, we propose a new BBR available rate calculation algorithm, BMR, which reduces max-filtered available bandwidth by multiplying the weight (less than $\frac{9}{10}$). Our evaluation results for BMR show that BMR can operate around the optimal operating point without increase of inflight packets.

Acknowledgments

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